

Where and Why We Introduce Novelty in the iRDRC Architecture

BTP Novelty Mapping Document

Built on the base paper: Hieu, Hoang, Luong, Niyato (IEEE WCL, Dec. 2020)

Abstract

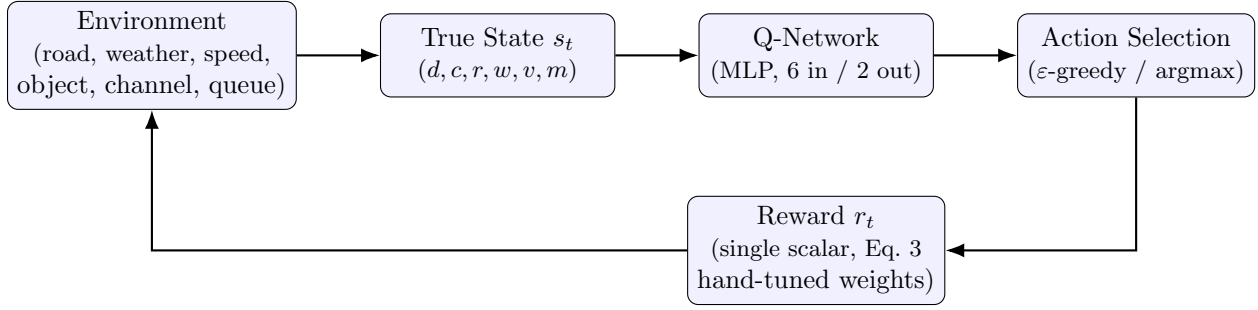
This document maps our two proposed novelty contributions directly onto the specific architectural components of the base iRDRC paper that they modify, and explains, in detail, why each modification targets that particular component rather than any other. It assumes the reader has already studied the section-by-section explanation of the base paper (companion document) and its consolidated list of simplifying assumptions – every modification proposed here is a direct response to one or more of those assumptions, not an arbitrary addition.

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1 The Base Architecture, as a Pipeline

Before locating our changes, it helps to see the base paper’s architecture as a pipeline of five components. Every one of our proposed changes attaches to exactly one or two of these five blocks – nothing else in the pipeline is touched.



#	Pipeline block	Base paper's assumption at this block
1	True state s_t	Assumed <i>perfectly observed</i> by the AV (Section III-A)
2	Q-network	Plain feed-forward MLP, takes raw state directly (Section IV)
3	Action selection	ϵ -greedy over a single unconstrained Q-value (Section IV)
4	Reward r_t	Single scalar, safety enforced only via a large hand-picked penalty weight $r_3 = 50$ (Section III-B)
5	Learning objective	Unconstrained maximization of expected discounted return

Our two novelty contributions modify blocks **1–2** (state observability) and blocks **4–5** (reward/objective) respectively. Block 3 changes only as a *consequence* of the other two, not as an independent change.

2 Novelty 1: From Perfect State Observation to Mixed-Observability (Belief-Based) Decision Making

2.1 The assumption being targeted

From the consolidated assumptions list: the base paper assumes all six state components (d, c, r, w, v, m) are known to the AV *exactly*, with no observation noise. We established earlier that this is unrealistic for at least two of these components:

- **Channel state c** – in the base paper, c is not sensed at all; it is drawn from a fixed probability ($p_c = 0.1$). In reality it would come from a noisy SINR/RSSI estimate, delayed ACK/NACK feedback, or a quantized CQI report.
- **Moving object state m** – detecting a hidden/emerging object (the paper's own example: a car obscured by a truck) is precisely the hard perception problem radar exists to solve. Treating m as a known binary flag sidesteps the actual detection uncertainty.

By contrast, d (the AV's own queue length) is exactly known internally, and v (speed, from the speedometer) and w/r (weather/road, from dedicated instruments) are comparatively low-noise readings in practice. Treating *all six* as equally uncertain would be its own kind of inaccuracy in the other direction.

Where exactly this modifies the base architecture

This modifies **Block 1 (True state $\rightarrow$ Observation)** and **Block 2 (Q-network architecture)** of the pipeline above – specifically:

- Section III-A’s state space definition, $\mathcal{S} = \{(d, c, r, w, v, m)\}$, is replaced with a **Mixed-Observability MDP (MOMDP)**: the state is split into a *fully observed* component $s_t^f = (d, r, v, w)$ and a *partially observed* component $s_t^p = (c, m)$, the latter accessed only through noisy observations $o_t \sim \Omega(o \mid s_t^p)$.
- Section IV’s Q-network (plain MLP, 6-input) is replaced with a **belief-augmented Q-network**: the input becomes (s_t^f, b_t) , where b_t is a belief vector over (c, m) , maintained either via a Bayes filter update or learned implicitly by a recurrent encoder (a DRQN-style GRU/LSTM layer) that consumes a short history of past observations and actions.

Why here, and why this matters

Why exactly here, and not elsewhere in the pipeline: the base paper’s observability assumption lives entirely in how state is *defined and fed into the network* (Blocks 1–2) – it has nothing to do with the reward function or the training objective. Fixing it anywhere else (e.g. just adding noise to the reward) would not actually address the problem; the AV genuinely needs a mechanism to reason about *uncertain* channel/object information, which requires changing what the network receives as input and how it processes a history of noisy readings, not just how it is rewarded.

Why the mixed-observability (MOMDP) framing specifically, rather than a full 6-dimensional POMDP: a full POMDP over all six variables is both unnecessary (four of them are genuinely low-noise in practice) and considerably harder to train and justify within a 10-week window. Restricting partial observability to (c, m) – the two variables that are honestly uncertain – keeps the belief-tracking problem low-dimensional and tractable, while still being a genuine, defensible departure from the base paper’s blanket full-observability assumption. This distinction (MOMDP vs. full POMDP) is itself a recognized, citable modeling choice in the POMDP literature (e.g. Ong et al., “POMDPs for Robotic Tasks with Mixed Observability”), not an ad-hoc simplification invented for convenience.

Why this is not already taken: our literature search found POMDP formulations applied broadly to autonomous driving (lane changing, navigation, obstacle avoidance), and separately found the transfer-learning and GNN-based follow-ups to iRDRC – but none of them apply any observability-aware reformulation to *this specific* radar/communication mode-selection problem.

Why it matters practically: this change is what makes a later real-hardware validation phase meaningful rather than cosmetic – real sensors genuinely produce noisy channel and object-presence readings, so a policy trained under the false assumption of perfect knowledge would be the *wrong* thing to deploy on real hardware in the first place.

3 Novelty 2: From Hand-Tuned Reward Shaping to a Formally Constrained Safety Objective

3.1 The assumption being targeted

From the consolidated assumptions list: safety in the base paper is enforced only by manually setting the miss-detection penalty $r_3 = 50$ large enough that the learned policy is *empirically* conservative. There is no bound, guarantee, or certificate that the resulting miss-detection probability stays below any specific, chosen threshold – it is whatever falls out of training with that particular weight.

Where exactly this modifies the base architecture

This modifies **Block 4 (Reward function, Eq. 3)** and **Block 5 (Learning objective)** of the pipeline – specifically:

- Section III-B’s single scalar reward r_t is **split into two separate signals**: a throughput reward r_t^{thr} (the original $+r_1/+r_2$ terms, with the $-r_3$ penalty removed), and a safety **cost** signal $c_t = \mathbb{1}[\text{miss detection at } t]$ (an indicator, not a hand-picked penalty magnitude).
- The learning objective in Section IV (unconstrained maximization of expected discounted return) is replaced with a **Constrained MDP (CMDP)** objective:

$$\max_{\pi} \mathbb{E} \left[\sum_t \gamma^t r_t^{\text{thr}} \right] \quad \text{subject to} \quad \mathbb{E} \left[\sum_t \gamma^t c_t \right] \leq \delta \quad (1)$$

for a chosen safety threshold δ , solved via a **Lagrangian-DQN**: a dual variable $\lambda \geq 0$ is introduced, actions are selected as $a_t = \arg \max_a [Q_r(s, a) - \lambda Q_c(s, a)]$, and λ is updated by dual ascent based on whether the constraint is currently being satisfied or violated – in place of the DQN update in Eq. 5 acting on one undifferentiated reward stream.

Why here, and why this matters

Why exactly here: the safety issue in the base paper is entirely a property of *how reward is defined and optimized* (Blocks 4–5) – it has nothing to do with whether the state is fully or partially observed. Even with perfect observability, the base paper’s reward design still offers no guarantee. This is a genuinely separate axis of improvement from Novelty 1, which is precisely why both can be introduced together without redundancy – they touch different, non-overlapping parts of the pipeline.

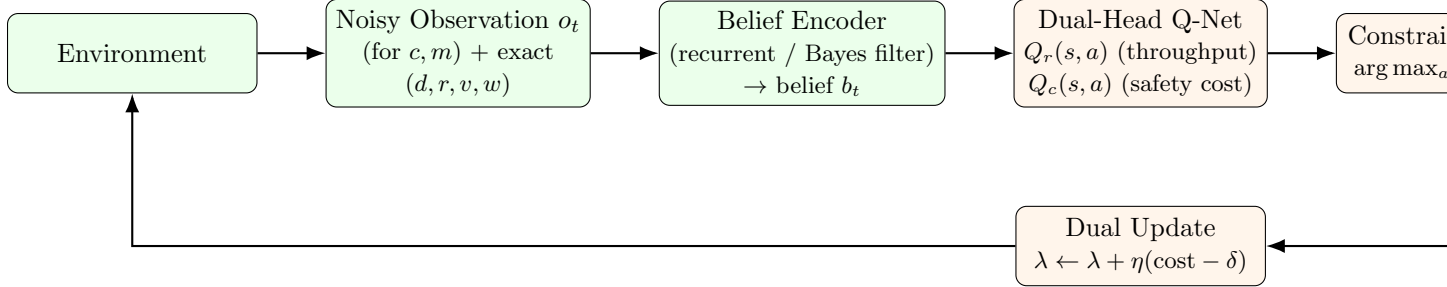
Why a constrained formulation rather than simply re-tuning the reward weights: re-tuning r_3 can shift the empirical safety/throughput trade-off, but it can never *guarantee* a specific bound – any fixed weight only produces a policy that happens to land somewhere on the trade-off curve. A constrained formulation lets us directly specify the acceptable miss-detection bound δ as a design parameter and search for the best throughput *subject to that bound being respected*, which is a fundamentally different (and stronger) guarantee than reward-shaping can offer.

Why this is not already taken: safety-constrained/Lagrangian RL is an active area in wireless systems generally (e.g. recent 2025–2026 work on constrained RL for URLLC uplink scheduling), but we found no application of a formally constrained objective to this specific radar/communication mode-selection safety problem – every iRDRC follow-up we found (transfer learning, GNN-based cooperation) keeps the original unconstrained, hand-weighted reward design unchanged.

Why this matters for publication: a reviewer in a safety-adjacent venue (ITS, vehicular networks, or wireless systems tracks with a safety angle) is likely to view “we can now provide a probabilistic bound on missed detections, not just a lower empirical rate” as a substantially stronger and more defensible claim than another percentage-point improvement on throughput.

4 How the Two Novelties Combine Into One Architecture

The two changes are independent in *where* they act (Blocks 1–2 vs. Blocks 4–5), but they combine into a single, coherent revised pipeline rather than two unrelated patches:



Green blocks correspond to Novelty 1 (observability); orange blocks correspond to Novelty 2 (constrained safety). Both feed into a single action-selection step, and both trace back to environment interaction through one loop – this is one unified reformulation of the base architecture, not two independent side-experiments bolted onto an otherwise unchanged pipeline.

Scoping note

Because the two novelties attach to non-overlapping parts of the pipeline, they are also **independently separable** if timeline pressure requires descopeing:

- Dropping Novelty 2 alone leaves a clean **MOMDP-only** contribution (belief-based mode selection under sensor uncertainty), still a valid, unclaimed improvement over the base paper.
- Dropping Novelty 1 alone leaves a clean **constrained-RL-only** contribution (formally safety-bounded mode selection), also independently valid and unclaimed.
- Keeping both gives the strongest, most “epic” single narrative – the base paper’s AV neither perceives its environment correctly nor guarantees the safety it claims to prioritize, and we fix both – at the cost of higher implementation and tuning effort within the 10-week window.

This document assumes the combined direction is the target unless we decide, after early implementation results, to descope to one half.

5 Summary Table

Architecture block	Base paper’s assumption	Our modification	Addresses assumption #
State definition (III-A)	All 6 state vars perfectly observed	Split into observed (d, r, v, w) + partially observed (c, m) via MOMDP	#5
Q-network (IV)	Plain MLP on raw state	Belief-augmented / recurrent encoder over noisy observations of (c, m)	#4, #5
Reward function (III-B)	Single scalar, safety via hand-picked $r_3 = 50$	Split into throughput reward r^{thr} + safety cost c_t	#7
Learning objective (IV)	Unconstrained max of expected return	Constrained MDP, Lagrangian-DQN with dual variable λ	#7
Action selection (IV)	$\arg \max_a Q(s, a)$	$\arg \max_a [Q_r(s, a) - \lambda Q_c(s, a)]$	— (consequence of above)

Assumption numbers refer to the consolidated list in the companion section-by-section explanation document.